Conditional generation of visible language

a Training

60 visible glyph images
Frozen causal reader

→

Continuous visual history
384 state + 1024 image field

History conditions every residual scale

Four noisy future cells
Independent or shared noise levels

→

Conditional strip denoiser
FiLM and cell attention

→

Velocity and visual losses
5.90M trainable parameters

b Inference

Noise and visible history

→

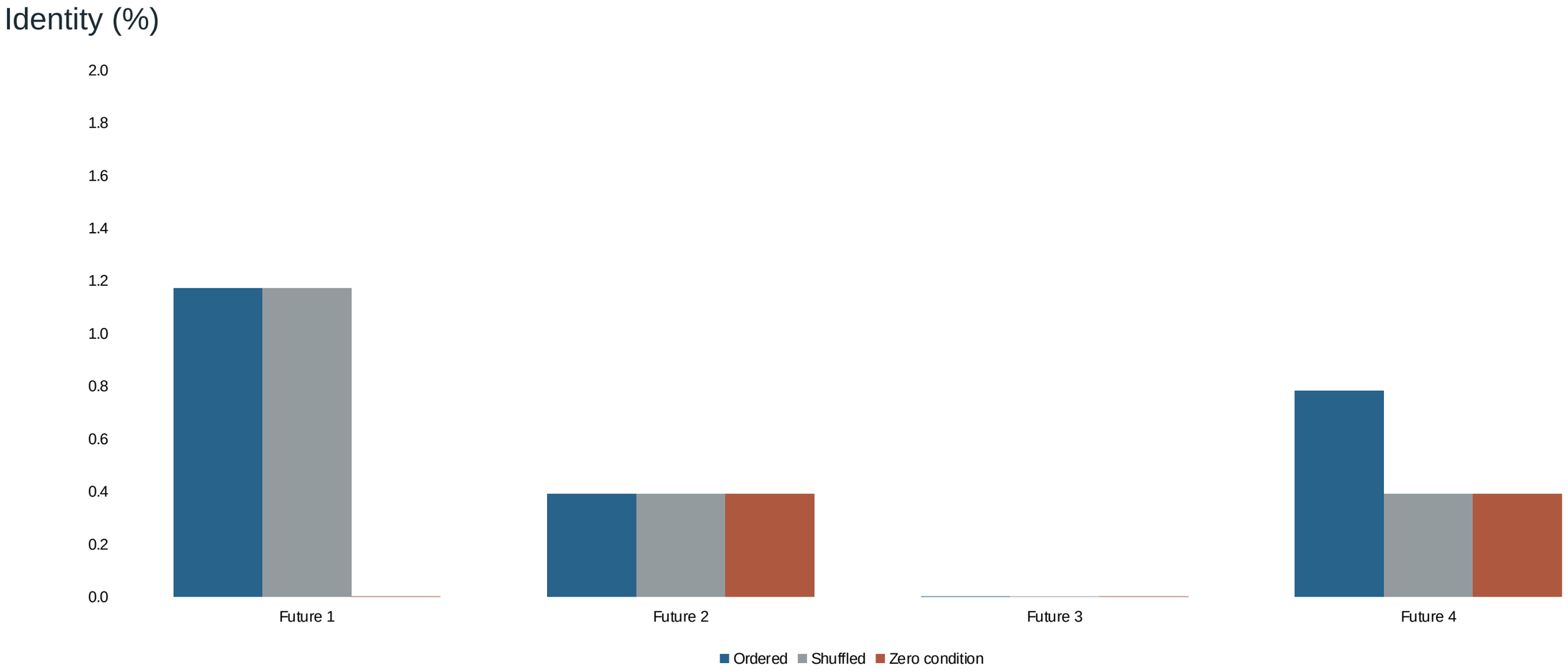
32-level reverse process

→

Four output images
Append pixels to next context

Source: V53 frozen protocol, implementation and locked audit. Schematic objects are editable.

Direct block generation under paired visual histories



256 prefixes, one primary sample each, four output cells
Ordered block identity 0.586%, shuffled 0.488%, zero condition 0.195%

Source: locked V53 development report. Native chart includes its editable evidence workbook.

Unselected output images expose the language error



Fixed first eight audit rows. Pixels are model output, with nearest-neighbour enlargement. No glyph replacement.

Exact raster likelihood preserves a visible copy path

64 writing images
Frozen 16.28M reader

→

Continuous visual state
Hidden384 + image field1024

→

Fixed inverse field
Aligned signed pixel plan

Training: exact Bernoulli image NLL

Earlier target rows
Four-layer row transformer

→

Earlier pixels within row
Strictly masked MADE

→

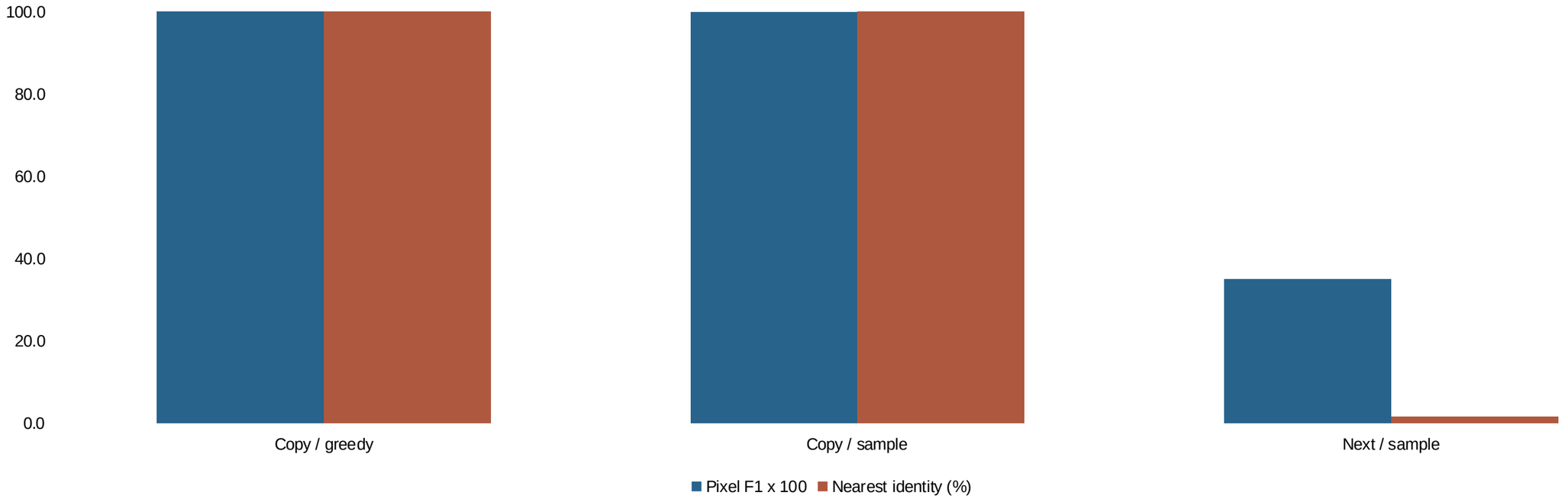
Aligned logit + residual
Normalized next-image law

25% supplied-image motor exercises; 75% natural predicted conditions. No glyph inventory.

Inference: sample 1,024 causal pixels; append the visible raster to history

2.43M new density parameters; 18.71M total. Supplied-image copying is architectural, not language evidence.

Copying succeeds; autonomous language remains unresolved



Exact-image copying: 256/256 greedy. Autonomous target identity: 4/256 ordered, 3/256 shuffled.

Target NLL: 15.219 ordered; 15.254 shuffled nats/image
Paired advantage 0.0355 nat; source-record bootstrap 95% CI [-0.1014, 0.1693]

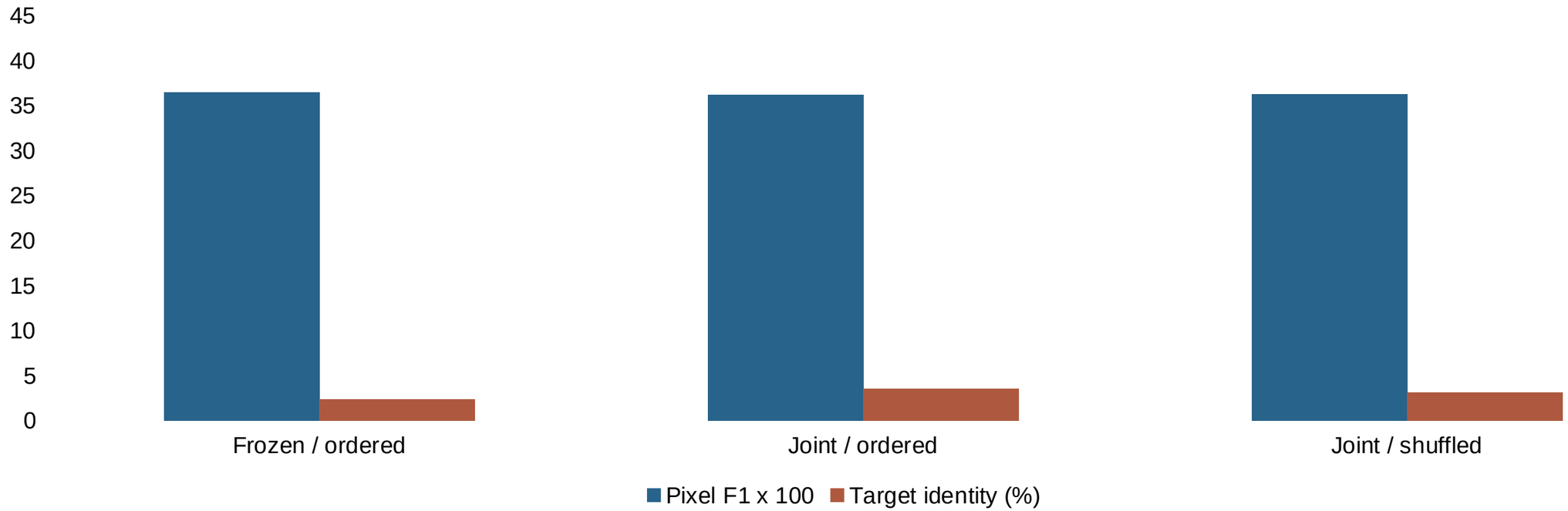
V54 locked development audit. Native chart and embedded workbook. Supplied-image controls reveal the answer.

Actual rasters: exact motor copying and failed continuation



First eight fixed audit rows; nearest-neighbour enlargement. No OCR, image replacement or selected examples.

Matched joint learning: better likelihood, incomplete generation

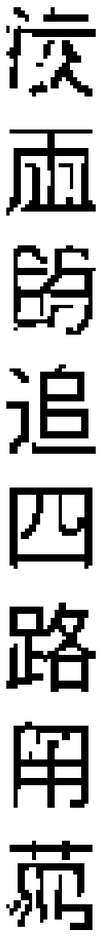
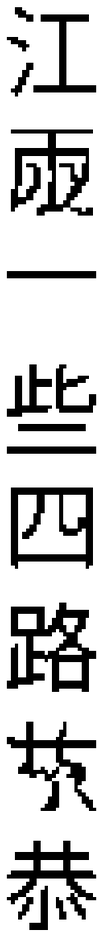
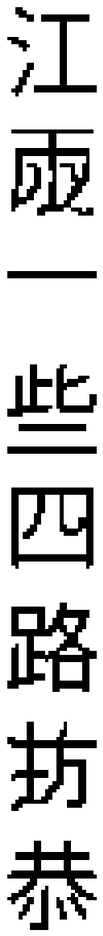


Exact target NLL: joint 7.026; matched frozen 7.579 nats/image
Paired gain 0.553; 95% source-record interval [0.450, 0.665]

Both arms retain exact supplied-image copying. Joint model passes 6/8 gates; generation remains unqualified.

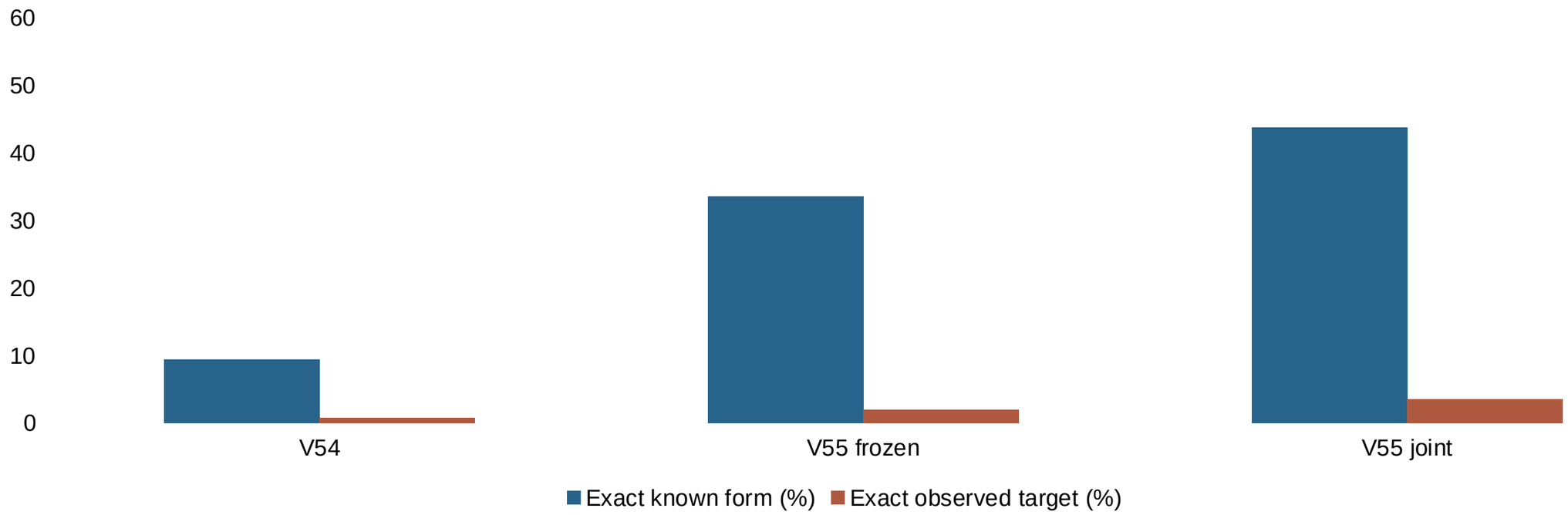
V55 locked matched audit; 512 NLL targets, 256 single-sample prefixes. Native chart includes its editable workbook.

Raw matched output: exact copying and remaining language errors

Target	Frozen sample	Joint sample	Joint shuffled	Joint greedy	Supplied copy
					

First eight fixed audit rows, no selection. Pixels are actual output; no OCR, evaluator-bank replacement or external inference.

Clear glyphs and correct language are separate tests

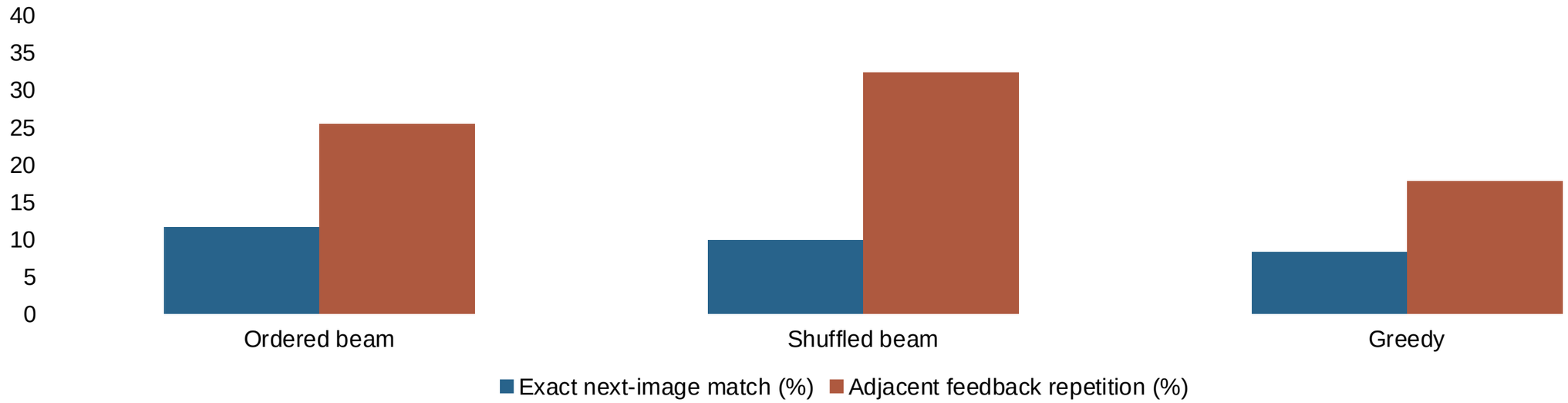


Exact known-form gain, joint minus frozen: 10.156 percentage points
95% source-record interval [4.219, 16.240]; 256 saved first samples

The fixed 9,932-image bank is used only for evaluation. No image is repaired or replaced.

Post-result diagnostic; all original qualification decisions remain unchanged. Native chart includes its editable workbook.

Raw-pixel beam search: better immediate choice, repetitive output

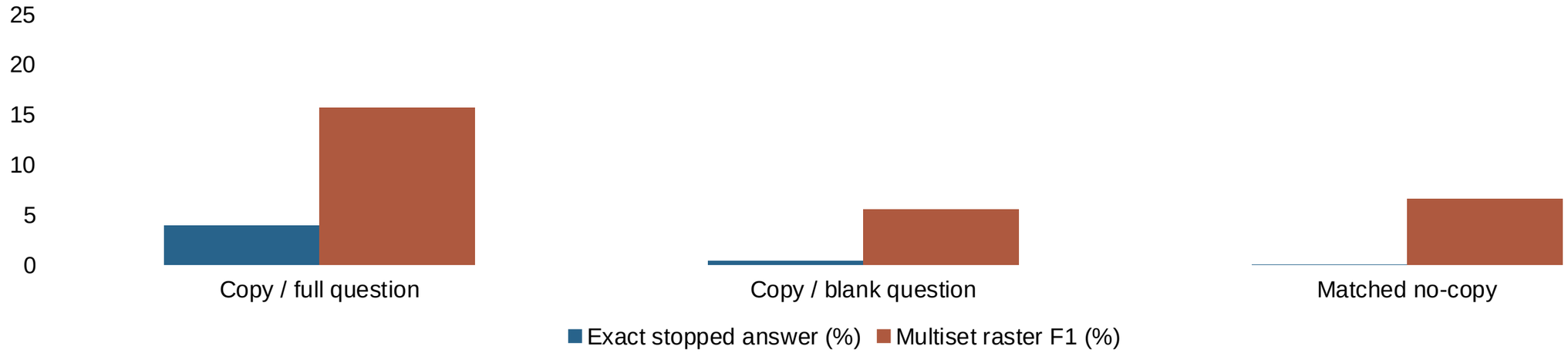


2,048 prefixes: beam gain over greedy 3.369 points [2.351, 4.412]; over shuffled 1.758 [0.966, 2.552]
Observed-stream repetition: 0.112%; beam: 25.446%. Rollout gate fails; decoder rejected.

Observed future	Ordered beam	Shuffled beam	Original greedy
不尬的就跑回来了	一个，不是一个，	一个，不不不不不	，不不一，不不不
的女婿。丈人教我	，不是一人，不是	，不不不不不不不	，不不不不不不不

Unselected first two eight-step audit streams. Unchanged V55 weights; no vocabulary or repair. Both controls use highest FP32 precision.

Natural Chinese questions: some binding, inaccurate complete answers



Question-dependent F1 gain: 10.165 points [7.043, 13.371]
Exact answers: 10/256; both questions correct: 1/393 pairs. Four of seven gates pass; model rejected.

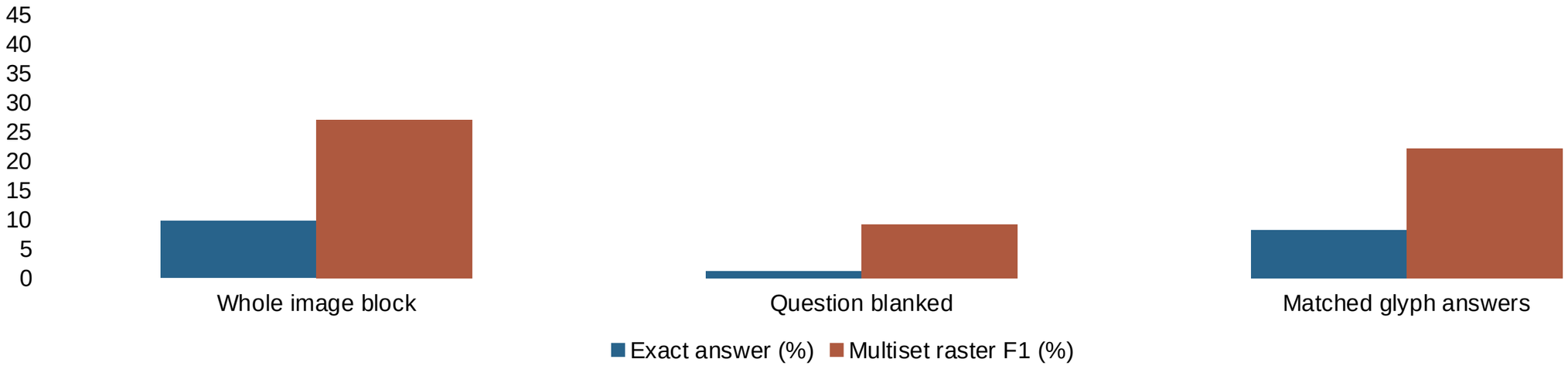
Q1 １９４０年８月世宗大学更名成什么？

Q2 邵伯温是在什么时候去世？

Reference 1	Copy	Blank question	No-copy
首尔女子师范学院 高宗绍兴四年（１１３４年）	１９４０年５月，Ｙ月世宗大学” 邵溥、邵博、邵传后》	节居绍圣、利州路转运作）	１９３０年 １３３０年

First two fixed audit rows; actual output images. White means zero emitted cells. All primary copy-arm answer cells use observed prompt images.

Whole observed image answers: better F1, unresolved exact-match gain



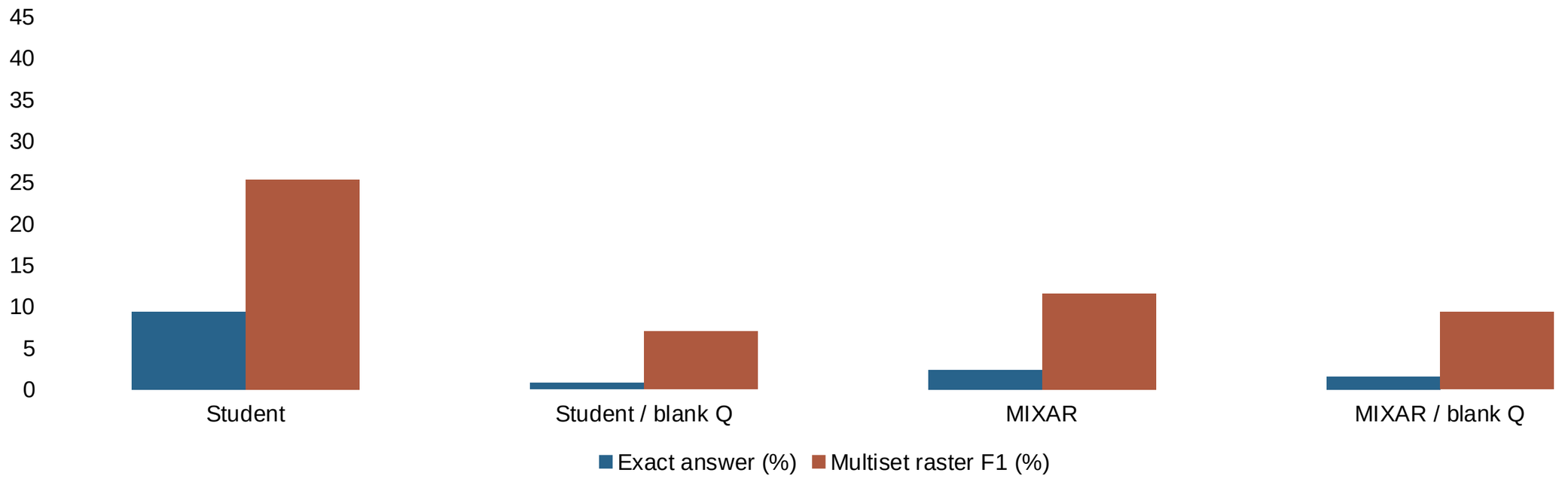
25/256 exact answers; matched glyph control 21/256. F1 gain 4.893 points [0.469, 9.047].
Only 4/393 paragraph pairs have both answers correct. Six of seven gates pass; reading rejected.

Prompt pixels → Causal reader → Start / end law → Source image answer

	Retained reference	Actual block answer
Audit row 1	首尔女子师范学院	“世宗学院”
Audit row 2	高宗绍兴四年（1134年）	卒于高宗绍兴四年（1134年）

First two fixed audit rows. Copying preserves selected pixels; it does not certify the meaning of the selected text.

Frozen visual foundation probe: larger pretrained features do not help here

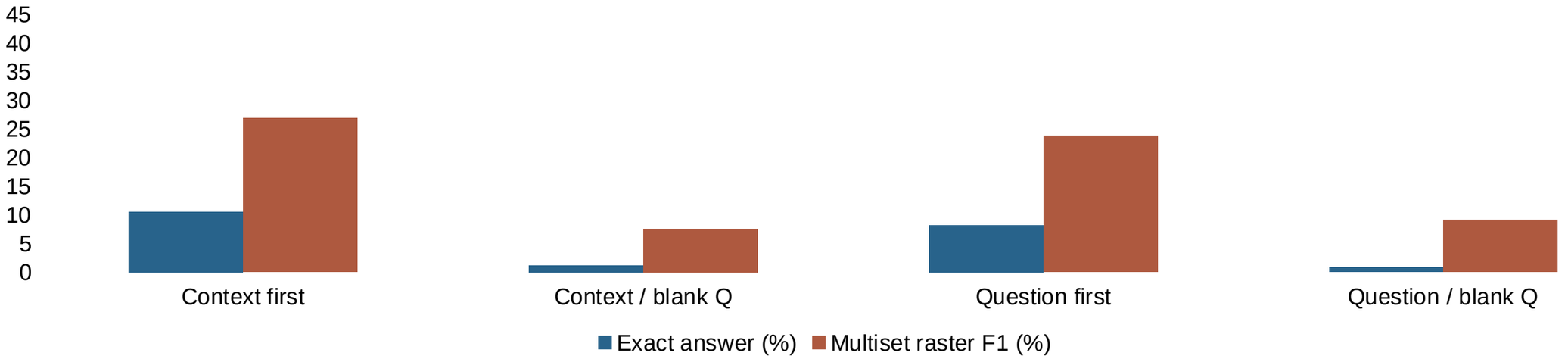


Identical 789,520-parameter heads; 4,000 matched updates each.
Student 24/256 exact; MIXAR 6/256. MIXAR-minus-student F1 −13.752 points [−18.247, −9.094].

MIXAR has different pretraining and native rendering; this probe uses our Noto CJK grid.
External four-H200 pretraining is not one-4090 work. Frozen readout comparison rejected; not a general encoder ranking.

Continuous prompt-image features only. Shared cached release weights; no external inference call or answer annotation enters feature preparation.

Does reading the question first improve evidence selection?



Context first

Passage images X → Question images Q → Answer

Question first

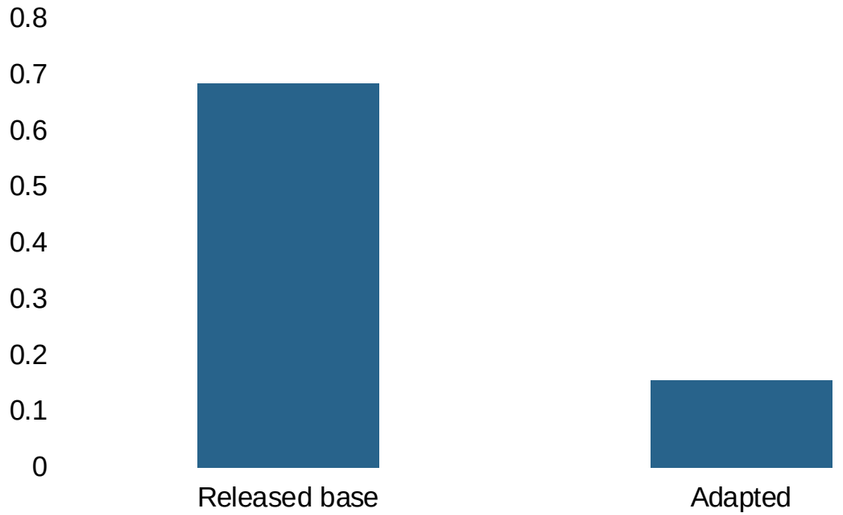
Question images Q → Passage images X → Answer

Question-first minus context-first F1: -3.143 points [-6.544, 0.446].
4/8 registered gates pass; decision: reject_question_first_intervention.

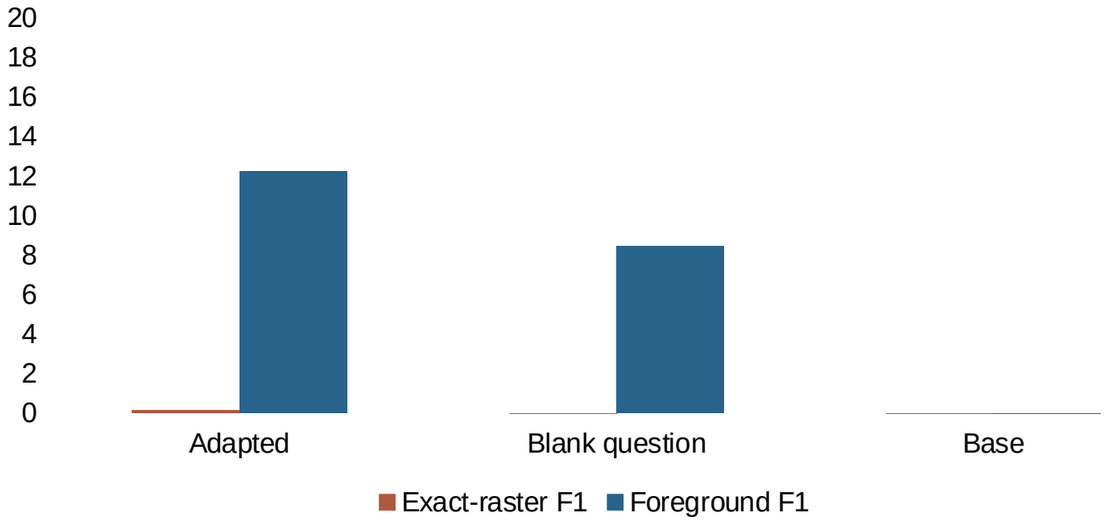
Same checkpoint, data, parameters and training budget. Source pixels only; reused development, not a sealed final test or general language qualification.

Native free writing: improved pixel fit, failed language generation

Held-out answer-pixel NLL (nats)



Generated image scores (%)



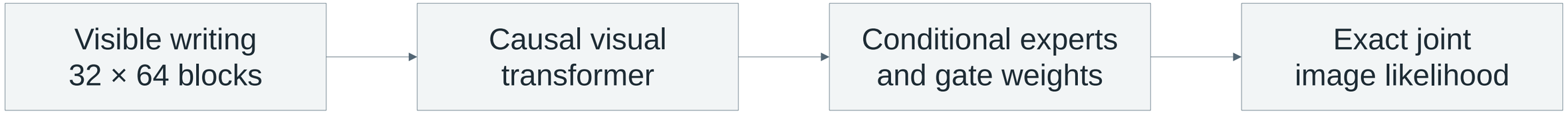
0/256 exact answers and 0/393 pairs both correct. Five of ten registered gates pass.
477.15M externally pretrained parameters, fully adapted for 1,000 updates on GPU 0.

	Retained reference	Actual free-pixel answer
Audit row 1	首尔女子师范学院	
Audit row 2	高宗绍兴四年（１１３４年）	

Training: 1,501.1 s and 9.912 GiB. Offline native-PNG replay passes. External pretraining and reused development disclosed.

Conditional whole-image density

Training: prompt and observed answer images



$$p(y \mid h) = \Sigma k\ \pi k(h)\ \Pi i\ \text{Bernoulli}(yi\ ;\ \text{sigmoid}(aki(h)))$$

One shared expert choice couples all pixels in a bitmap. Each expert is a neural function of context.
K = 1: 477.15M parameters. K = 8: 490.00M. Same examples and update budget; different head size.

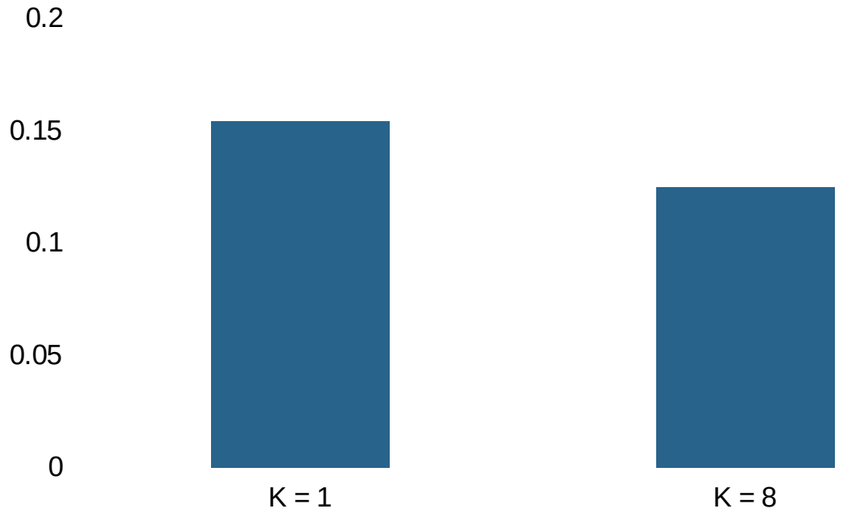
Inference: image prompt → candidate bitmaps → score under full density → physical pixels

Feed generated pixels back into the visual context; stop at an all-white image block.
Component-mode search is approximate MAP. No character bank, OCR, Unicode IDs or external LM in the actor.

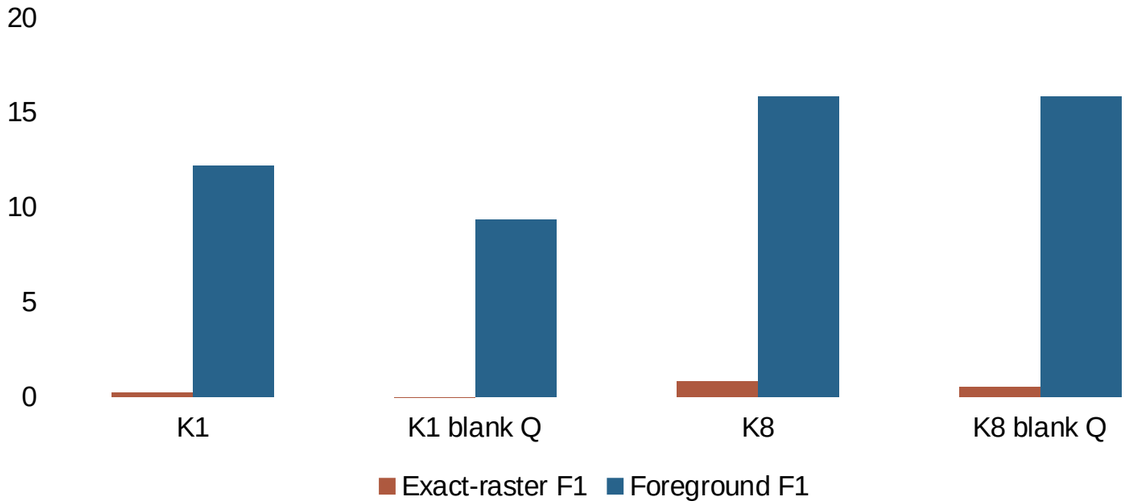
Externally pretrained MIXAR foundation. The mixture law is a tested hypothesis, not a guarantee of readable language.

Conditional density: held-out fit and generated writing

Held-out answer-pixel NLL (nats)



Generated image scores (%)



Exact answers K1 / K8: 0 / 0 of 256. Both-correct pairs: 0 / 0 of 393.
Registered decision: reject density intervention. General image-language quality remains unqualified.

	Retained reference	K = 1	K = 8
Audit row 1	首尔女子师范学院	[illegible]	1 9 4 0 年
Audit row 2	高宗绍兴四年（1 1 3 4 年）	[illegible]	1 9 [illegible]

First two fixed primary answers, unrepaired; white display padding. One seed, reused development, external pretraining disclosed.

Question-conditioned image-language training

a Observed training images: two questions about one paragraph

Question region Q0

哪个人是世纪越南反唐起义军的军事首领？

Recorded answer Y0 冯兴

Question region Q1

冯兴与他的弟弟冯骇接受谁的建议，率众起义？

Recorded answer Y1 杜英翰

Each full prompt Xi also contains the same paragraph image. Appended white termination blocks are not drawn.

b Four causal likelihoods under the same parameters

(X0, Y0) (X0, Y1)
(X1, Y0) (X1, Y1)

→

Shared image generator
490M parameters, 8 experts

→

	Y0	Y1
X0	e_{00}	e_{01}
X1	e_{10}	e_{11}

eij = mean negative log likelihood of the complete answer suffix, including termination

c Positive image likelihood with paired question assignment

$$D = e_{01} + e_{10} - e_{00} - e_{11}$$

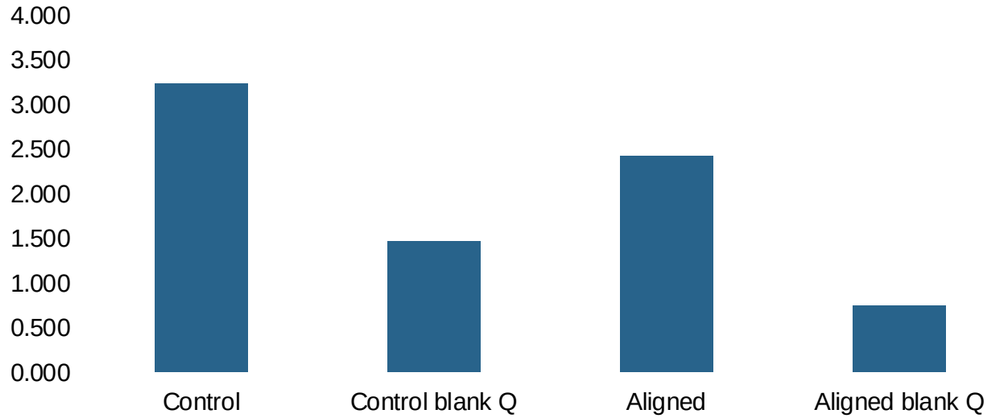
λ = 0: control; λ = 1: assignment

$$L = (e_{00} + e_{11}) / 2 + \lambda \cdot 0.01 \cdot \mathrm{softplus}(-D / 0.01)$$

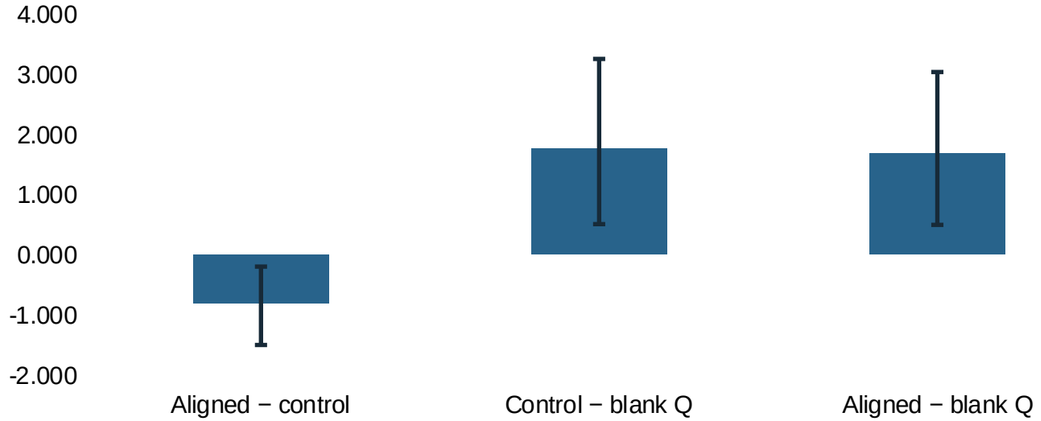
Observed targets, not model outputs. Both arms start from the same V62 EMA; autonomous answers are evaluated separately.

Question assignment: completed free-writing test

Strict-raster F1 (%)



Paired differences with 95% intervals (pp)



Exact answers: control 2/256, aligned 1/256. Both-correct pairs: 0/393 in each arm.

First two fixed primary answers

1: Reference 首尔女子师范学院

1: Control



1: Aligned



2: Reference 高宗绍兴四年（1134年）

2: Control



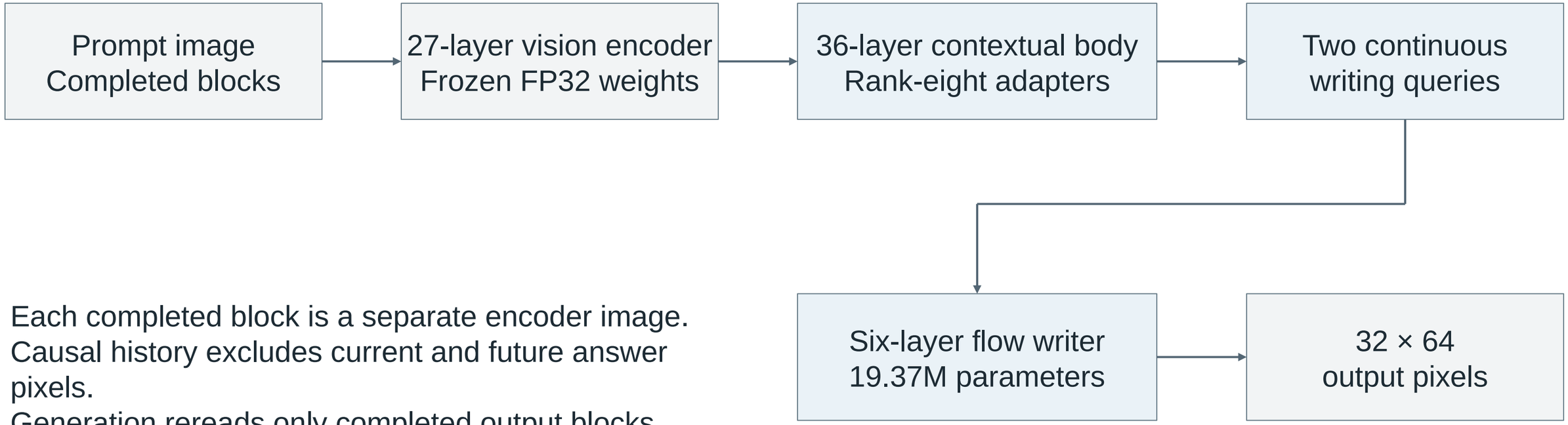
2: Aligned



Unrepaired pixels and white padding. One seed, reused development and external pretraining. Both reading gates fail.

Complete continuous image actor

7.548B total parameters 25.58M trainable 17.181 GiB peak allocated



Each completed block is a separate encoder image.
Causal history excludes current and future answer pixels.
Generation rereads only completed output blocks.

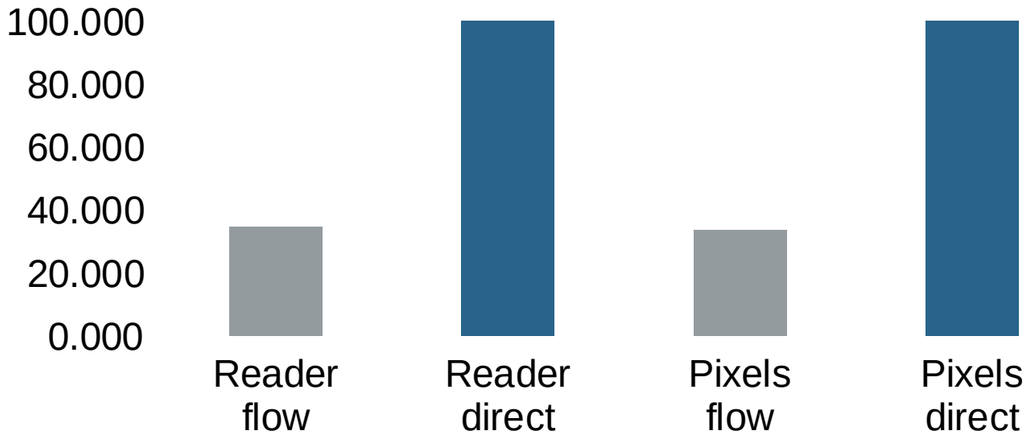
$$Yt = (1 - t)\,U + t\,Y \qquad U \sim N(0, I) \qquad Lflow = E\,||v(Yt,t,h) - (Y - U)||^2$$

Frozen contextual weights use BF16 storage with FP32 computation. A separate classifier predicts stopping.

Execution checks pass. Writing and language quality remain unqualified.

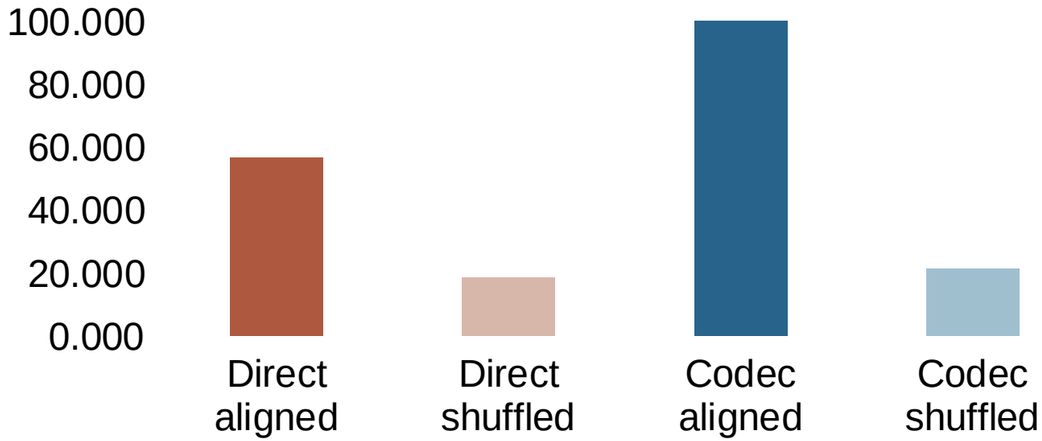
Writing fidelity depends on the output representation

a Training F1 (%) | n=64



Exact blocks: 0/64, 63/64, 0/64, 64/64

b Development F1 (%) | n=128



Exact aligned: direct 3/128; fixed codec 126/128

c First two fitting rows

	Reference	Reader flow	Reader direct	Pixels flow	Pixels direct
1					
2					

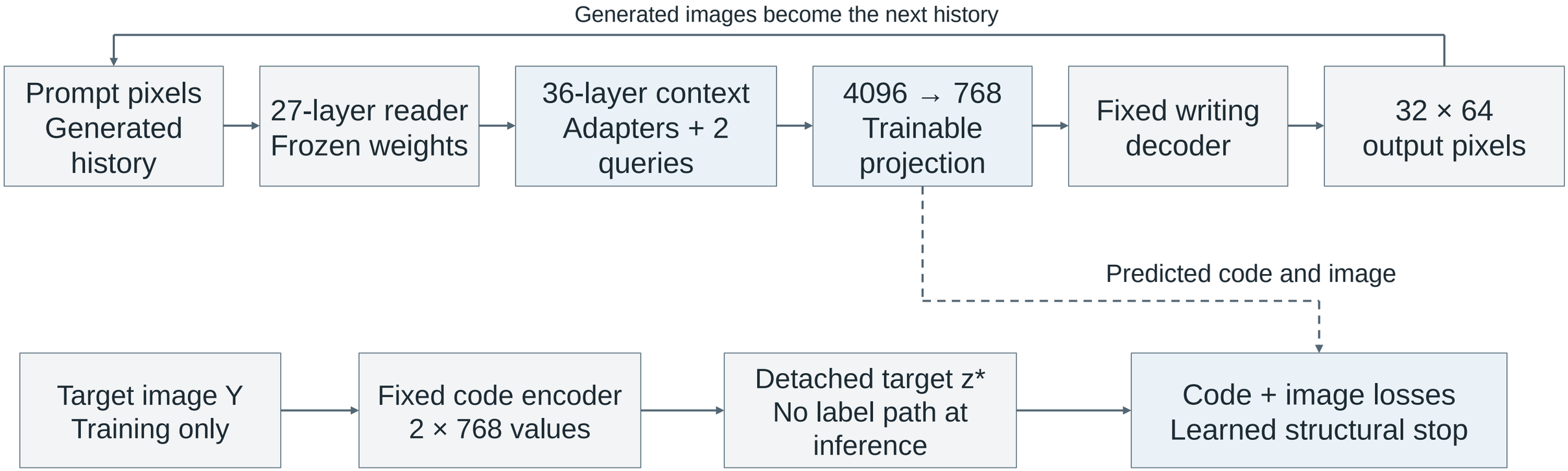
d First two development rows

	Reference	Direct	Codec
1			
2			

Fitting uses training images. Development compares supplied representations with unequal prior training; neither is question answering.

Reading and writing use separate continuous spaces

7.539B total parameters 9.352M trainable 16.953 GiB peak allocated



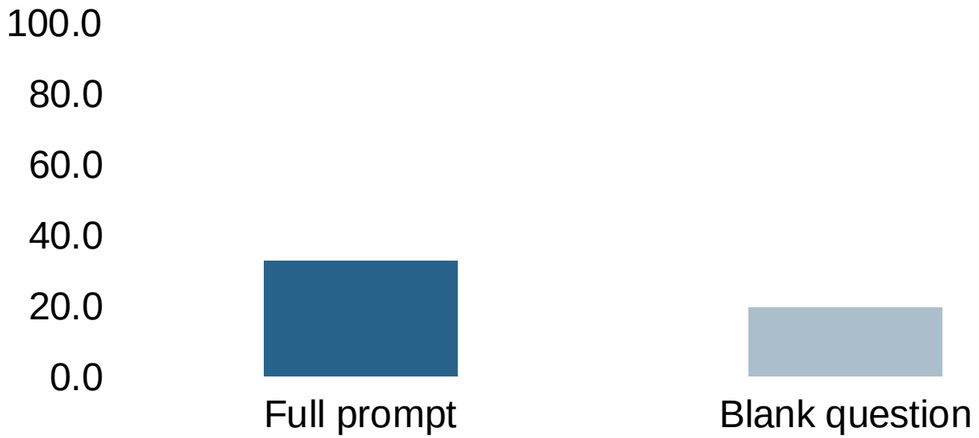
$$L = MSE(N(\hat{z}), N(z^*)) + MSE(D(\hat{z}), Y) + Lstop$$

Frozen decoder weights still pass gradients into the predicted codes and contextual adapters.

All 24 complete-model execution checks pass. Two-update output is fragmented; autonomous language remains unqualified.

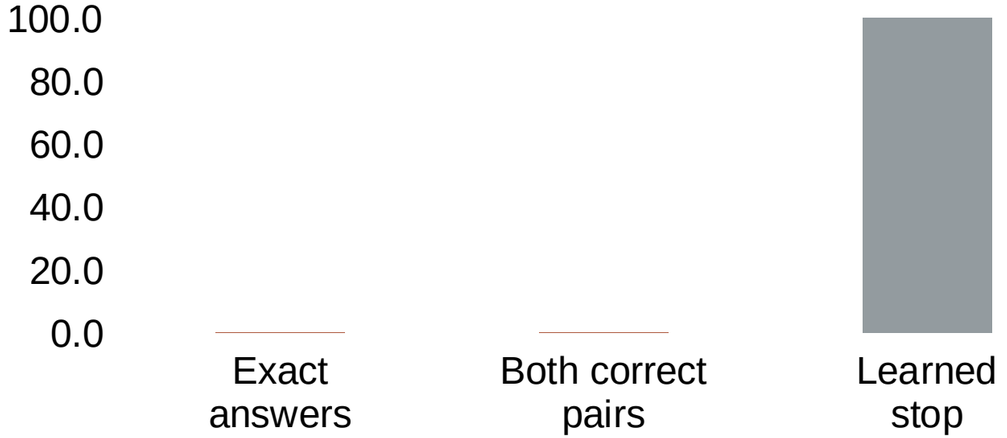
Question effects persist when accurate writing fails

a Chinese foreground F1 (%) | n=128



Paired gain: 13.16 points [10.93, 15.73]

b Exact answers and learned stopping (%)



0/128 exact answers; 0/64 complete pairs; 128/128 stops

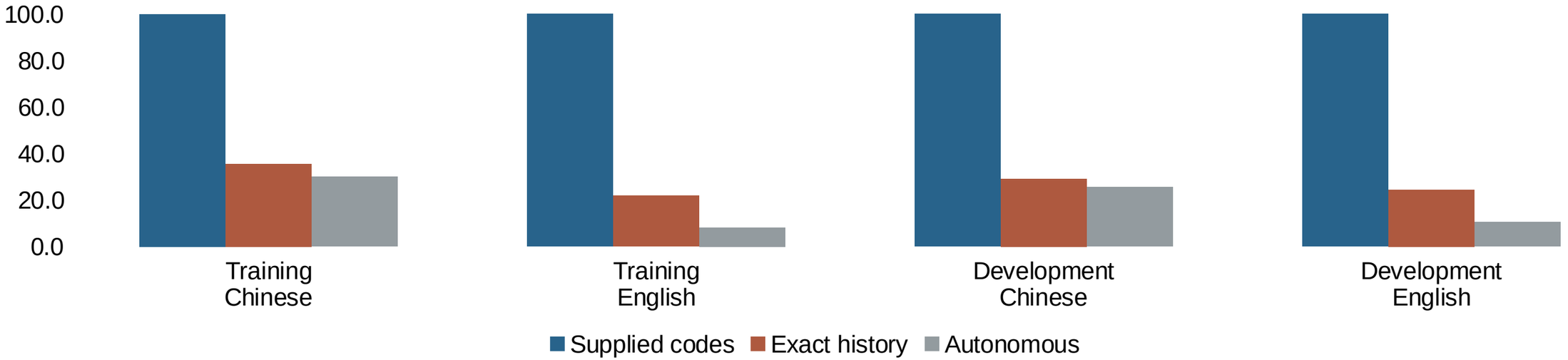
c First two fixed Chinese cases | complete, unmodified streams

	Reference	Full prompt	Blank question
1	玛丽莲·曼森	[illegible]	[illegible]
2	批评他的人	[illegible]	[illegible]

All 384 generations and original gates are verified. Question dependence and stopping pass; answer quality fails.

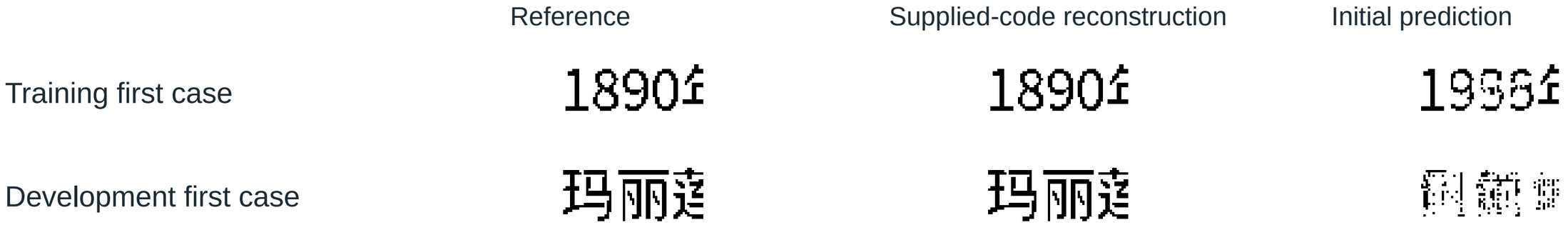
Prediction fails before generated-history drift

a Foreground F1 (%) | eight fixed cases per group



Exact history supplies previous images and answer length; it is an assisted diagnostic.

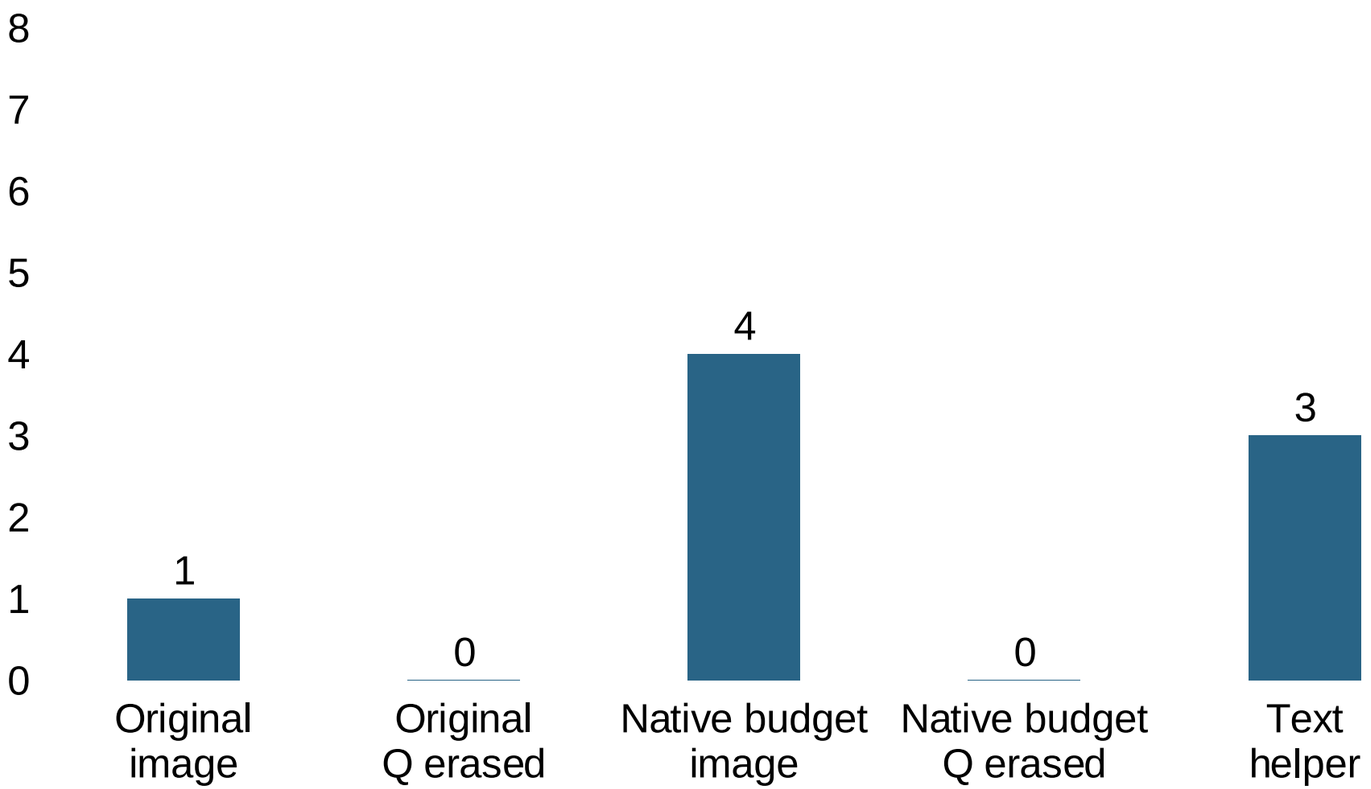
b First 32-by-64 block, before generated history



No first predicted block is exact. All 32 initial predictions match the autonomous path; no diagnostic weights update.

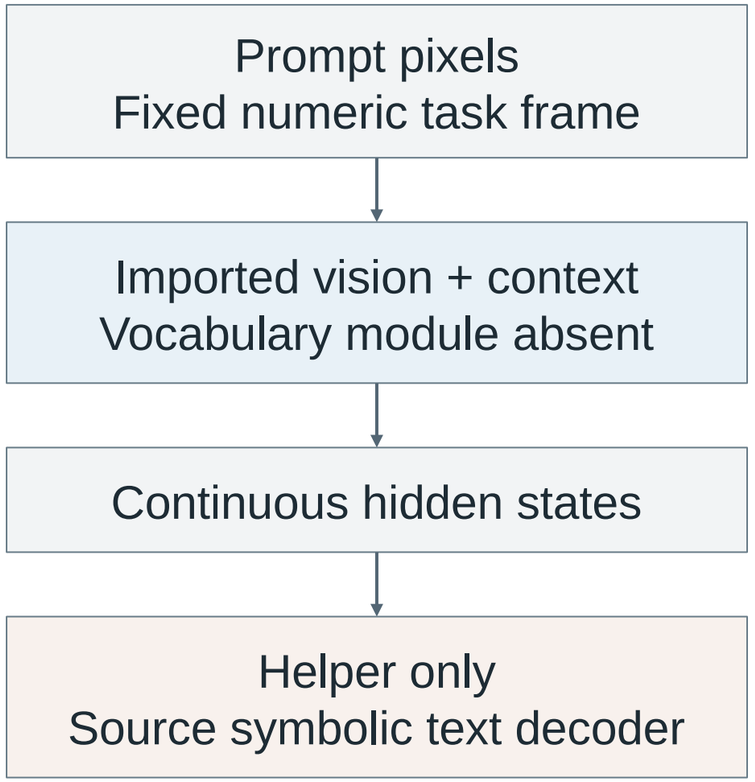
Numeric image controls retain bounded reading

a Exact Chinese answers | the same eight reused questions



Q erased: only question ink is removed, with wrap coordinates retained.

b Diagnostic boundary

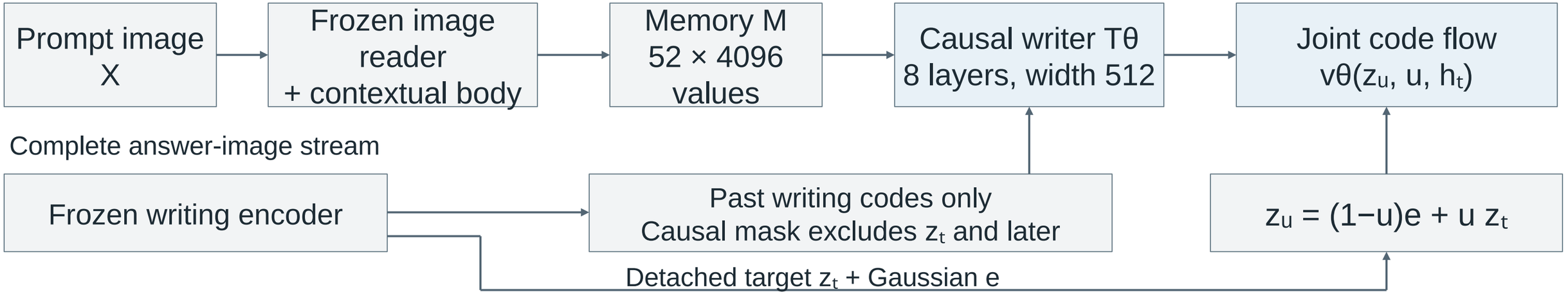


Text decoding is excluded from the image actor.
This check generates no image answers.

All 32 input images and 742 retained decoding choices verify. The complete neural forward pass is not independently replayed.

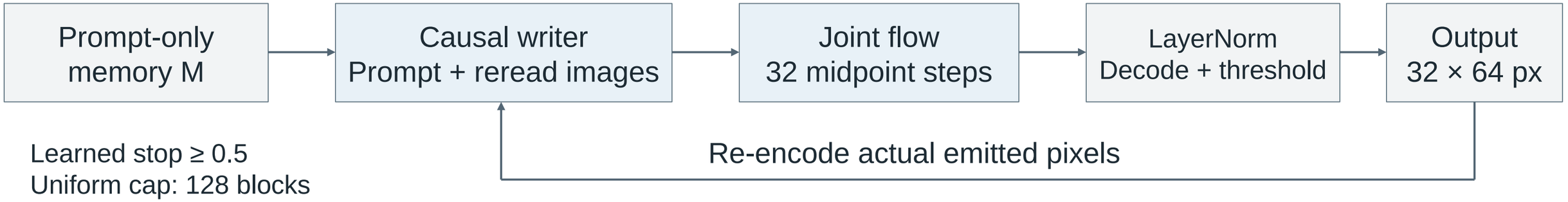
Causal image writing with frozen visual semantics

a Training | complete answer images supervise continuous writing



$$L = \text{mean } ||v\theta - (z_t - e)||^2 + \text{balanced stop loss}$$

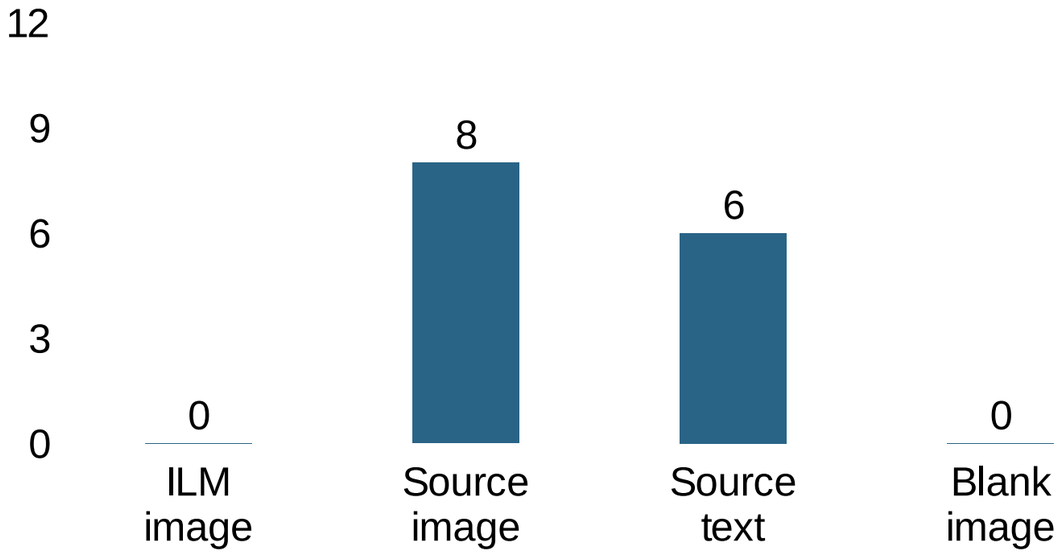
b Inference | every generated block becomes physical visual history



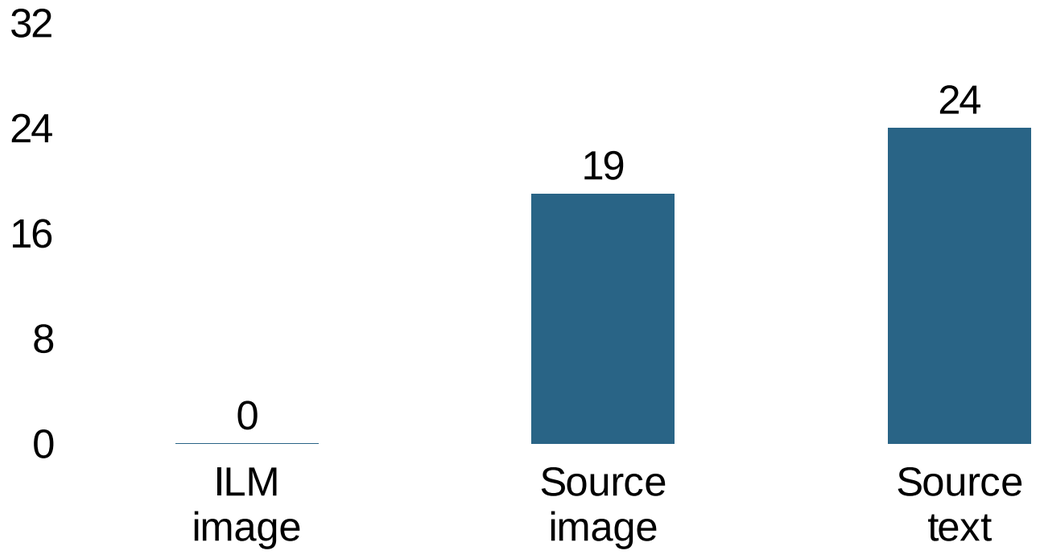
38.65M trainable writer. External reader pretraining retained. Completed Chinese audit: 0/128 exact; 29.386% foreground F1.

Reader content before image writing

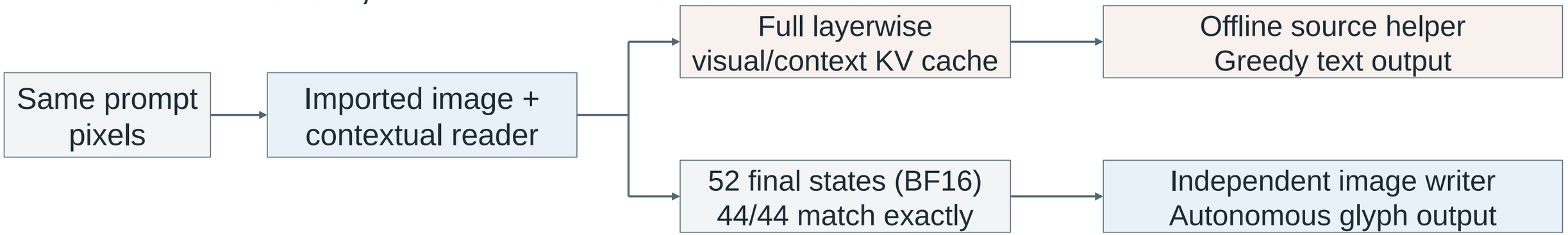
a Exact copied answers / 12 new requests



b Exact answers / 32 fixed training questions



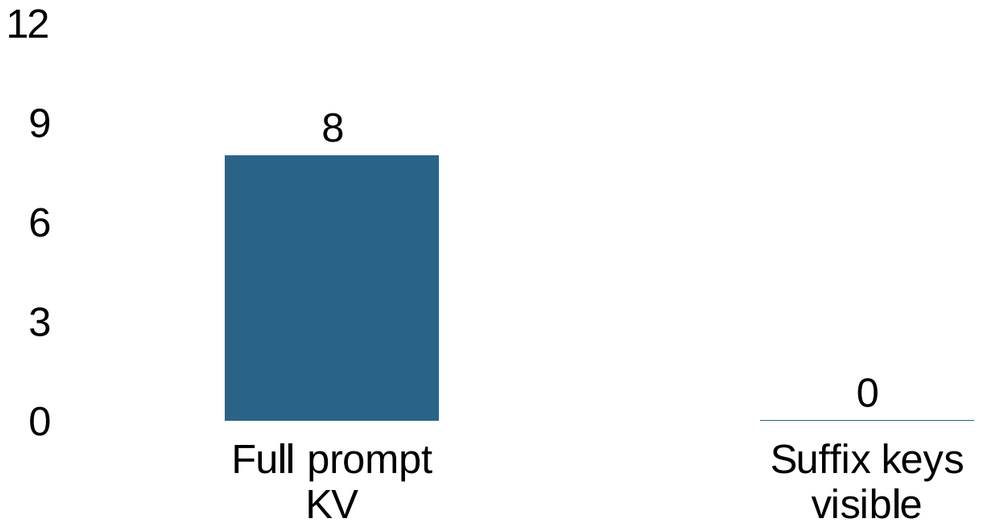
c Shared observation; different continuation information



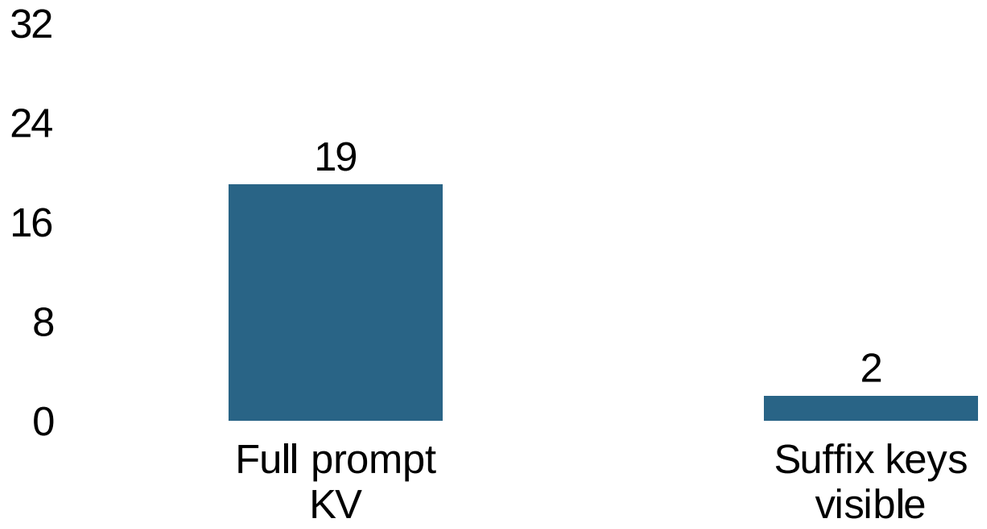
100 fixed offline helper calls. Full-KV success does not prove compact-memory sufficiency or successful ILM image generation.

Context visibility and native image continuation

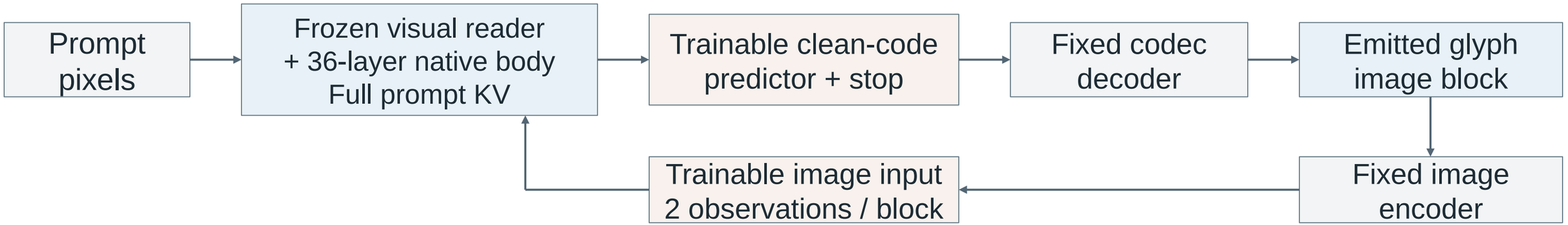
a Exact copy answers / 12 requests



b Exact answers / 32 training questions



c New image actor: execution only; no trained language result



7.583B total parameters; 46.900M trainable interface parameters. External pretraining retained; 16.638 GiB execution peak.